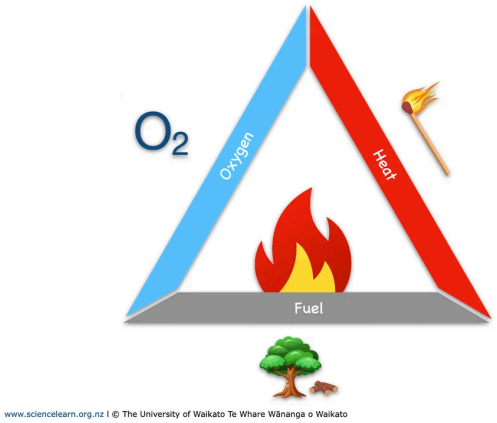
What is fire?

Fire is the visible effect of the process of **combustion** – a special type of chemical reaction. It occurs between oxygen in the air and some sort of fuel. The products from the chemical reaction are completely different from the starting material.

The fuel must be heated to its ignition temperature for combustion to occur. The reaction will keep going as long as there is enough heat, fuel and oxygen. This is known as the fire triangle.



Combustion is when fuel reacts with oxygen to release heat energy. Combustion can be slow or fast depending on the amount of oxygen available. Combustion that results in a flame is very fast and is called burning. Combustion can only occur between gases.

Fire is composed of three elements, namely **heat**, **fuel**, and **oxygen**. These three elements form the fire triangle. A fire starts when fuel is heated from an external source and reaches its ignition temperature. Oxygen reacts with the fuel to sustain the fire. Once fire has started, it will continue to burn until an element of the fire triangle is removed.

What are the common fire hazards?

Many fires start due to negligence such as leaving the stove unattended or by not organizing the storage cabinet. The things that we find useful in our homes such as paper, paint, cooking oil, and electrical appliances can ignite and burn easily. The following are common causes of fire:

1. Faulty electrical wiring or connection	9. Neglected electrical appliances or devices
2. LPG-related	10. Unchecked electrical types of machinery
3. Neglected cooking or stove	11. Flammable chemicals
4. Lighted cigarette butts	12. Incendiary device or ignited flammable liquid
5. Left unattended open flame: torch or sulo	13. Spontaneous combustion
6. Left unattended open flame: candle or gasera	14. Under planned pyrotechnics
7. Matchstick or lighter kept near flammable materials	15. Bomb explosion
8. Direct flame contact or static electricity	16. Lightning



Flammable materials could also ignite on their own in an event referred to as spontaneous combustion. This happens when an object produces its own heat without absorbing heat from its surroundings.

When it reaches a temperature enough for ignition, it will start a fire by itself. Everyone should be mindful of materials that are prone to spontaneous combustion such as spilled oil, oily rags, paper, dried leaves or hay. It is better to store them in well-ventilated areas or outdoors to prevent ignition and to easily dissipate the heat.

What is Fire classification?

There are different 5 type of fire, and each type has different property and characteristic, It is important to know what kind of fire you are dealing with since different types of it can occur, and depending on the type you have to take different measures to take when you encounter them.

- **Class A**: Fires involving ordinary combustible materials such as wood, cloth, paper, rubber, and many plastics.
- **Class B**: Fires involving flammable and combustible liquids and gases, like gasoline, oil, and propane.
- **Class C**: Fires involving energized electrical equipment, such as wiring, motors, and appliances.
- **Class D**: Fires involving combustible metals like magnesium, sodium, and potassium.
- **Class K:** Fires in cooking appliances that involve combustible cooking media, such as vegetable or animal oils and fats.